

Wording, Response Options, Order, and Sensitive Questions

FROM A MEASUREMENT INSTRUMENT TO A RESPONDENT'S ANSWER

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Last week: What are we trying to measure?

Today: What happens when a **human respondent**
encounters our measure?

Where we stopped

Last week we moved from:

Research objective → construct → dimensions → indicators

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Today we finish the right-hand side of the chain:

indicators → question → respondent process → observed response

Continuity, not recap

We are not starting a second topic. We are opening up the mechanism that turns a question into data.

Finish the reading workshop

Group 1: Groves Ch. 7

What model of survey measurement is being proposed? Where can error enter?

Group 2: Krosnick & Presser

What principles govern the move from a construct to a question and response format?

Group 3: Pew

How are those principles translated into professional questionnaire practice?

Small-group work: 8 minutes

Return to the reading with your group. Your job is to teach us the **argument**, not summarize pages.

Four questions for every reading

Each group should be ready to answer:

1. What is the central **methodological claim**?
2. What assumption about respondents or measurement is underneath it?
3. What **design decision** follows from that claim?
4. What kind of error could result if we ignore it?

Your turn

When another group reports, prepare one challenge: *When might that recommendation fail, or create a different problem?*

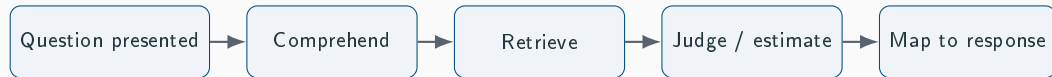
Graduate methods should not end with “best practices.”

- Does the recommendation depend on the **construct**?
- Does it depend on the **population**?
- Does it depend on **mode**?
- What is the tradeoff between respondent burden and measurement precision?
- Could reducing one source of error increase another?

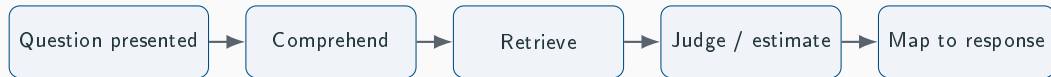
The standard

A design rule is useful only if you understand the mechanism it is trying to control.

What Groves adds to the measurement chain



What Groves adds to the measurement chain



Each stage can introduce error — even if the respondent is cooperative and trying to answer well.

Groves et al. (2009), Ch. 7; Tourangeau, Rips, and Rasinski (2000).

A perfect sample does not solve measurement error

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of the **wrong quantity**.

Why?

- the construct was underspecified;
- the item captures only one dimension;
- respondents interpret the question differently;
- retrieval is difficult;
- the response options distort mapping;
- context changes the judgment being reported.

Open and closed questions: same construct?

Open

What do you think is the most important problem facing the country today?

Closed

Which of the following is the most important problem facing the country today?

- Cost of living
- Immigration
- Health care
- Crime
- Education
- Something else

Your turn

Are these two instruments measuring the same thing? Defend your answer.

Open versus closed is a measurement choice

	Open-ended	Closed-ended
Cognitive task	Recall / generate	Recognize / select
Researcher control	Low	High
Respondent burden	Often higher	Often lower
Standardization	Lower	Higher
Risk	Coding + articulation	Omission + framing
What becomes salient?	Respondent-generated	Researcher-provided

Not a commandment

Choose the format that best matches the inferential task and the respondent's cognitive task.

Response options are part of the measure

Imagine asking:

How often have you felt very irritated during the past month?

Two researchers provide very different frequency ranges.

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Why might the **range itself** alter the answer?

Bridge to Schwarz

Respondents can use response alternatives as information about what the researcher means, what counts as typical, and which distinctions matter.

The questionnaire is not a neutral container.

It is part of the **stimulus** respondents are reacting to.

Schwarz: how questions shape answers

This week's readings

1. **Schwarz (1999)** — cognitive and communicative processes underlying self-reports.
2. **Pew Methods 101 (2018)** — practical consequences for wording and ordering.
3. **Tourangeau & Yan (2007)** — what changes when respondents may not want to report truthfully.

The common problem

The observed answer depends not only on the respondent's underlying state, but on the task we create for them.

Self-reports of attitudes and behaviors are shaped by features of the research instrument — including wording, format, and context.

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The important part is not merely that “wording matters.”

The effects arise from **systematic cognitive and communicative processes**.

Respondents answer more than the literal sentence

Consider:

“How often do you exercise regularly?”

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“How often do you exercise regularly?”

The respondent must infer:

- What counts as **exercise**?
- What does **regularly** mean?
- Which time period is relevant?
- What level of precision is expected?
- Why is the researcher asking this question here?

Literal + pragmatic meaning

Survey answering is a constrained conversation. Respondents infer the researcher's intended meaning from the words and the surrounding instrument.

The questionnaire as a conversation

Schwarz draws on the logic of ordinary conversation: respondents generally assume that the researcher is trying to be meaningful and informative.

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That means features we may treat as “formatting” can be interpreted as **information**:

- response categories;
- numeric endpoints;
- examples included in the stem;
- reference periods;
- questions that came immediately before;
- what the questionnaire appears to be about.

Your turn

What information might a respondent infer from a scale that ranges from “0 times” to “more than 10 times”?

Behavior reports: retrieval or estimation?

Suppose I ask:

“How many times have you eaten at a restaurant in the past six months?”

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Are you retrieving six months of discrete episodes from memory?

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Are you retrieving six months of discrete episodes from memory?

Probably not. For frequent, mundane behaviors, respondents often construct an estimate using:

- typical rates (“about twice a week”);
- partial retrieval;
- category-based inference;
- cues provided by the question and response options.

Compare:

1. How many times did you eat at a restaurant during the past seven days?
2. How many times did you eat at a restaurant during the past six months?

Reference periods change the cognitive task

Compare:

1. How many times did you eat at a restaurant during the past seven days?
2. How many times did you eat at a restaurant during the past six months?

Same behavior. Different retrieval problem.

Design implication

A reference period is not just a calendar boundary. It determines whether respondents can retrieve events or must estimate them.

Group A: Comprehension

“How often have you exercised regularly during the past year?”

Diagnose literal and pragmatic ambiguity.

Group B: Retrieval

Compare a 7-day and 6-month restaurant-use question.

Explain how the response strategy changes.

Group C: Response scale

Design two different frequency scales for the same behavior.

Predict how each scale could alter reports.

Small-group work: 10 minutes

Do not merely rewrite the question. Identify the **cognitive or communicative mechanism** that creates the problem.

For each diagnosis, we want four pieces:

1. What is the respondent trying to infer or retrieve?
2. What feature of the instrument changes that task?
3. What direction of error or heterogeneity might result?
4. What redesign would reduce the problem — and what might that redesign sacrifice?

Graduate-level diagnosis

“Bad wording” is not a mechanism. Tell me **why** the observed answer changes.

Response scales can communicate meaning

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Therefore, two scales with the same verbal question can induce different interpretations of:

- what counts as “often”;
- what counts as “normal”;
- which distinctions the researcher wants the respondent to make.

Question order can change the judgment itself

Suppose respondents answer:

1. How satisfied are you with your dating life?
2. How satisfied are you with your life as a whole?

versus the reverse order.

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The later answer may reflect a different **information set**, not a different person.

Context effects are not always “more of the same.”

Assimilation

Previously activated information is **included** in the target judgment.

Example logic: “My dating life is good, so my life feels good.”

Contrast

Previously activated information becomes a **comparison standard** or is excluded from the target.

Example logic: “Compared with that unusually good case, this seems worse.”

Why order effects are substantive

Question order can change what respondents treat as evidence for the judgment.

Order experiment: predict before we test

Imagine a survey contains these two modules:

- detailed questions about personal financial strain;
- overall evaluation of one's quality of life.

Your turn

Would you expect the quality-of-life measure to differ if the financial-strain module comes first? Under what theory?

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Possible mechanisms:

- accessibility / priming;
- conversational inference about what “quality of life” is supposed to mean;
- assimilation or contrast;
- fatigue or satisficing — a different mechanism entirely.

Survey respondents do not simply “reveal” an answer.
They **construct** an answer under the conditions we design.

**Pew: translating mechanisms into
design decisions**

Pew turns theory into production rules

Pew emphasizes that sound questionnaire design requires attention to:

- clear, specific, answerable wording;
- exhaustive and mutually exclusive categories;
- question order and context;
- wording and context when measuring change over time;
- mode when interpreting trends;
- social desirability for sensitive behaviors and attitudes.

Ask the deeper question

What failure in the response process is each practice trying to prevent?

Map the “rule” to the mechanism

Design practice	Measurement problem it is trying to control
Clear, specific wording	Heterogeneous comprehension
Concrete reference period	Retrieval / estimation error
Exhaustive categories	Forced misclassification
Non-overlapping categories	Ambiguous response mapping
Attention to order	Context-dependent judgment
Same wording for trends	Changes in the measurement instrument
Self-administration for sensitive items	Social desirability / disclosure pressure

A wording change can change the estimand

Pew's historical example asks about support for U.S. military action in Iraq in 2003.

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Your turn

Is one version “biased”? Or are they asking respondents to evaluate different policy scenarios?

A wording difference can reveal several different things:

1. one version is genuinely confusing or leading;
2. the versions activate different considerations;
3. the versions define different hypothetical conditions;
4. the construct itself is conditional on information respondents receive.

Not every wording effect is a defect

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Analytic discipline

Before calling a wording effect “bias,” specify the construct and the estimand each version is supposed to measure.

Trend measurement creates a different problem

Suppose an item has been asked identically for 15 years, but today you believe you can write it better.

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Suppose an item has been asked identically for 15 years, but today you believe you can write it better.

Two goals now conflict:

Improve current measurement

- clearer wording
- better categories
- new mode / device realities

Preserve comparability

- same wording
- same context
- same response task

Measurement invariance problem

A better item today may break the time series you care about.

Small-group design decision: fix it or preserve it?

A client has a long-running question:

“How would you rate the state of your household finances today: excellent, good, only fair, or poor?”

The client wants to add explicit language about “paying bills and handling unexpected expenses” because they think respondents interpret “finances” inconsistently.

Small-group work: 8 minutes

Choose a recommendation. Keep the old item, replace it, run both, or do something else. State what evidence you would want before changing the trend.

Debrief: there is no free redesign

A defensible transition strategy might involve:

- a split-ballot bridge experiment;
- running old and new measures in parallel for one or more waves;
- testing mode-by-wording interactions;
- documenting a series break rather than disguising it;
- deciding explicitly whether the priority is **level estimation** or **change over time**.

Production lesson

Questionnaire design includes preserving the meaning of a measure across versions of the instrument.

**Tourangeau and Yan: when
respondents may edit the truth**

Sensitive questions are not one category

Tourangeau and Yan distinguish several ways a question can be sensitive.

1. **Intrusiveness**: the topic itself feels private or inappropriate to ask.
2. **Threat of disclosure**: truthful reporting could have consequences if disclosed.
3. **Social desirability**: one response is normatively more acceptable than another.

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Your turn

Is household income sensitive for the same reason as illegal drug use? What about voting?

Tourangeau and Yan (2007), pp. 859–870.

Sensitivity is relational and situational

A question is not simply “sensitive” in the abstract.

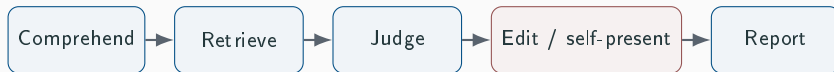
Sensitivity can depend on:

- what the respondent's true answer is;
- who appears to be asking;
- whether an interviewer is present;
- who else is nearby;
- perceived confidentiality;
- legal, social, or interpersonal consequences;
- norms within the respondent's social context.

Tourangeau & Yan's important move

Misreporting is often better understood as a **situational response strategy** than as a stable tendency of “dishonest respondents.”

Sensitive questions add an editing stage



For sensitive items, respondents can know the truthful answer perfectly well and still choose not to report it.

Tourangeau and Yan (2007), conclusion.

Sensitivity can produce different kinds of missingness and error

Response	What happened?
Unit nonresponse	Person declines the survey, perhaps because of the topic.
Item nonresponse	Person participates but refuses / skips the sensitive item.
Misreporting	Person answers, but edits in a socially safer direction.
Heaping / categorization	Person provides a less precise answer that feels safer.
Mode selection / breakoff	Person changes engagement or leaves when privacy feels insufficient.

Why this matters

Item response is not evidence that measurement succeeded.

Directional error is especially dangerous

Ordinary reporting errors can sometimes add noise in both directions.

Sensitive-question editing often has a directional component:

- socially desirable behavior may be **overreported**;
- socially undesirable behavior may be **underreported**;
- stigmatized attitudes may be softened;
- threatening disclosures may be concealed.

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- socially desirable behavior may be **overreported**;
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- stigmatized attitudes may be softened;
- threatening disclosures may be concealed.

That creates **bias in estimates**, not merely larger variance.

Design features can change reporting

The evidence reviewed by Tourangeau and Yan implies that reporting is responsive to survey design.

Common levers include:

- self-administration rather than interviewer administration;
- physical / social privacy;
- credible confidentiality protections;
- response formats that reduce the interpersonal act of disclosure;
- wording that lowers unnecessary judgment or threat;
- specialized techniques for highly sensitive behaviors.

Important caution

No technique makes sensitivity disappear. Each intervention changes the response task and may introduce new costs.

Your client wants an accurate measure of household financial strain.

They want to ask about:

- annual household income;
- missed housing or utility payments;
- debt burden;
- receipt of public assistance;
- inability to pay for necessities.

Small-group work: 12 minutes

Design a short module that you would be willing to defend methodologically and ethically.

Your group must specify:

1. Which items are sensitive — and why?
2. Where do you expect item nonresponse versus misreporting?
3. Would interviewer presence change the problem?
4. Exact values or categories? Why?
5. What introduction, privacy cue, or response design would you use?
6. What information are you willing to sacrifice to reduce respondent burden or threat?
7. How would you know whether your redesign actually improved measurement?

Debrief: “more disclosure” is not the only objective

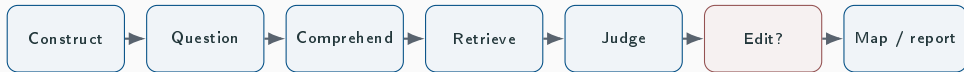
A strong design has to balance at least four goals:

1. **Validity**: does the response correspond to the target construct?
2. **Accuracy**: are respondents reporting what actually happened?
3. **Respondent protection**: are we minimizing unnecessary risk and intrusion?
4. **Usability**: can the resulting data support the intended analysis?

Ethics and measurement are linked

A design that makes respondents feel unsafe can be both ethically weaker and methodologically worse.

One integrated model



- **Groves:** distinguish constructs, measurement, response, and error.
- **Schwarz:** the instrument changes comprehension, retrieval, judgment, and mapping.
- **Tourangeau & Yan:** sensitivity can add motivated editing before reporting.
- **Pew:** professional design choices try to control these mechanisms in practice.

For your own survey work, every proposed item should survive five questions:

1. What construct / dimension is this intended to measure?
2. What cognitive task does the respondent need to perform?
3. What information does the wording, scale, and context communicate?
4. Is there a reason the respondent might edit or withhold the answer?
5. What claim will we make from the resulting variable?

The production standard

If you cannot answer these five questions, the item is not ready for fielding.

“Avoid double-barreled questions.”

“Use neutral wording.”

“Put sensitive questions at the end.”

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“Put sensitive questions at the end.”

Those can be useful heuristics. But the methodological goal is to understand:

what response process the design creates, for whom, and with what consequences.

Four ideas to leave with

1. A response is **constructed**, not simply retrieved from a mental file.
2. Wording, response alternatives, and order can change the **meaning of the task**.
3. Sensitive questions create not only nonresponse but **motivated reporting error**.
4. Good questionnaire design is the design of the **conditions under which measurement occurs**.

You are handed a survey item that seems clear and grammatically clean.

Your turn

What are the **three most important questions** you would now ask before deciding that it is a good measure?

Next: cognitive testing and multilingual equivalence — how we find out whether respondents are actually doing what we think they are doing.

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